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ARTICLE

Outcomes of a low birth weight phenotype on piglet gut microbial composition and intestinal transcriptomic profile

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Abstract: Decades of selection for increased litter size has caused a proportion of sows to consistently produce low birth weight (LBW) litters resulting in economic loss for producers due to reduced piglet survivability and growth. We hypothesized that piglets from LBW litters would have altered gut microbial composition, intestinal architecture, and intestinal transcriptomic profiles compared with piglets from high birth weight (HBW) litters. Sows were designated LBW ($n = 45$) or HBW ($n = 46$) based on litter birth weights of three successive parities. LBW piglets were 22% lighter ($P < 0.001$) at birth; however, no longer differed ($P > 0.05$) in weight at weaning compared with HBW piglets. LBW piglets had reduced ($P < 0.05$) fecal microbial diversity with a 114% increase in fecal *Enterobacteriaceae* ($P < 0.05$), as well as reduced ($P < 0.05$) abundance of cecal *Roseburia* and *Faecalibacterium*, fiber-degrading butyrate producers. Several genes associated with metabolic (*PER2*, *CES1*, *KLHL38*, and *HK2*) and immune pathways (*IL-1B*, *IRF8*, and *TNIP3*) were differentially expressed, suggesting altered metabolic and immune function in LBW piglets. In conclusion, LBW piglets had potentially unfavorable shifts in microbial structure in comparison to HBW piglets accompanied with alterations in metabolic and immune gene expression. Results indicate some biological consequences linking LBW phenotype to changes in production efficiency later in life.

Key words: piglet, microbiota, transcriptomics, low birth weight.

Résumé : Des décennies de sélection pour une augmentation de taille des portées ont causé une proportion des truies à produire de façon constante des portées à faible poids à la naissance (LBW — « low birth weight »), ce qui se traduit par une perte économique pour les producteurs imputable à la réduction de la survie et de la croissance des porcelets. Nous avons émis l'hypothèse que les porcelets des portées LBW ont une composition microbienne intestinale, architecture intestinale, et profils transcriptomiques intestinaux altérés par rapport aux porcelets de portées de grands poids à la naissance (HBW — « high birth weight »). Les truies ont été désignées LBW ($n = 45$) ou HBW ($n = 46$) selon les poids des portées à la naissance de trois parités successives. Les porcelets LBW étaient plus légers de 22 % ($P < 0{,}001$) à la naissance; mais ils ne différaient plus ($P > 0{,}05$) en poids au sevrage par rapport aux porcelets HBW. Les porcelets LBW avaient une diversité microbienne fécale réduite ($P < 0{,}05$) avec une augmentation de 114 % d'*Enterobacteriaceae* fécale ($P < 0{,}05$) ainsi qu'une réduction ($P < 0{,}05$) de l'abondance des bactéries *Roseburia* et *Faecalibacterium* cæcales, productrices de butyrate dégradant les fibres. Plusieurs gènes associés aux voies métaboliques (*PER2*, *CES1*, *KLHL38* et *HK2*) et immunitaires (*IL-1B*, *IRF8* et *TNIP3*) étaient exprimés de façon différentielle, suggérant des altérations dans les fonctions métaboliques et immunitaires chez les porcelets LBW. En conclusion, les porcelets LBW avaient des changements potentiellement défavorables dans la structure microbienne par rapport aux porcelets HBW, qui étaient accompagnés d'altérations dans l'expression de gènes métaboliques et immunitaires. Les résultats indiquent certaines conséquences biologiques associées au phénotype LBW liées aux changements d'efficacité de production plus tard dans la vie. [Traduit par la Rédaction]

Mots-clés : porcelet, microbiote, transcriptomique, faible poids à la naissance.

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Introduction

In the modern day swine industry, piglet health and efficiency of production are of utmost importance and can be severely reduced by high variation in piglet birth weights, with a low individual birth weight (LBW) being linked to increased percentage of body fat in the carcass (Krueger et al. 2014) and increased days to market (Smit et al. 2013). LBW piglets can ultimately come from one of two categories (a) within-litter LBW piglets (<1.0 kg in an average birth weight litter) or (b) between-litter LBW piglets (entire litters of LBW piglets) (Foxcroft et al. 2009). In many litters, a proportion of offspring are born in the bottom quartile for birth weight, and these runt pigs typically have poor viability and struggle for resources amongst larger littermates. However, decades of direct selection for increased litter size has indirectly increased variation in average birth weights between-litters, ultimately reducing growth efficiency and sustainability of production in a growing population of LBW litters (Gondret et al. 2005; Foxcroft et al. 2009).

A subpopulation of these more prolific sows have been observed to consistently have LBW litters of piglets regardless of litter size and have been designated LBW phenotype sows. This LBW phenotype is hypothesized to be driven by imbalances between key reproductive traits (ovulation rate and embryonic survival) that result in excessive intra-uterine crowding in early gestation, lasting negative consequences for placental development, and resultant intra-uterine growth restriction (IUGR) of all fetuses surviving to term (Foxcroft et al. 2009). Uterine crowding in early gestation, during placentation, affects placental development ultimately reducing uterine capacity and its ability to support fetal development throughout the gestational period (Stenhouse et al. 2019). Regardless of source, LBW piglets typically have low viability, impaired growth performance, poor carcass quality due to alterations in adipose and lean tissue deposition, and they are challenged immunologically (Quiniou et al. 2002; Poore and Fowden 2004; Town et al. 2004; Rehfeldt and Kuhn 2006; Foxcroft et al. 2009; Alvarenga et al. 2013). The poor growth performance of LBW piglets may be due to impaired or delayed gut maturation (Michiels et al. 2013), reduced ileal density (weight per length), villous area (D'Inca et al. 2011), and mucosal height (villous height plus crypt depth) (Alvarenga et al. 2013), ultimately reducing nutrient absorption and digestive efficiency. Evidence that neonatal stillborn pigs from sows with a LBW phenotype have a brain to intestinal weight ratio indicative of IUGR and lower wet placental weights at farrowing (Smit et al. 2013), underpins the concept of prenatal programming of postnatal performance as an inherent issue for LBW litters. Established in humans (Ferguson 1978) and pigs (Zhong et al. 2012), IUGR is associated with cellular immune impairment including impaired mucosal immunity (Dong et al. 2014), which may lead to greater risk of infection-related death. Interestingly, piglets from LBW phenotype sows (between-litter LBW pigs) may not have the same immune impairment as seen in within-litter LBW pigs, as no changes were observed in between-litter LBW pigs immune response after influenza A challenge (Wilkinson et al. 2015).

To date, most research efforts in LBW piglets have focused on those arising from within-litter variation in birth weight, with relatively little research on piglets from LBW phenotype sows. Although genetic selection has sought to reverse the trend for increased litter size to be negatively associated with litter average birth weight, by using co-selection for both numbers born and litter average birth weight, a better understanding of the developmental competence and lifetime consequences of pigs born from LBW phenotype sows is critical for improving the overall efficiency of pork production. The fundamental role colonizing gut microbiota play in the development and function of the intestine and immune system is well understood. Therefore, research establishing this relationship in piglets from LBW phenotype sows will help to explain the reduced efficiencies observed. We hypothesized that piglets from LBW phenotype sows would have altered gut microbial composition, intestinal epithelial architecture, and intestinal transcriptome when compared with piglets from high birth weight (HBW) phenotype sows.

Materials and Methods

Animals and phenotype

This study was performed in accordance to the guidelines of Pork Quality Assurance Plus® and Holden Farms Inc. ethical guidelines and with approval of the Faculty Animal Care and Use Committee-Livestock, University of Alberta (Edmonton, AB, Canada). The study was performed on a commercial multiplier farm ($n = 2400$ sows, Holden Farms, Inc.) located near Northfield, MN, USA. The study was initiated in March 2014. A total of 91 Camborough multiparous sows were assigned to trial at parity 2 or 3 and housed and managed according to barn protocols through an average of 5.8 ± 0.1 parities. Metrics of sow productivity recorded for successive parities included total born, total born alive, number of mummies and stillborn, and average litter birth weight. Sows were designated to have HBW or LBW phenotype litters based on their average litter birth weight over three successive parities. To classify birth weight phenotype, the relationship between total born (litter size) and litter average birth weight (average of all pigs in the litter) across the population and within parity was determined. The average litter birth weight for individual litters (total pigs born) within sow and parity was then compared with the fitted line, and within total litter size; the litter was categorized as H or L based on whether it fell approximately one standard deviation above or below the average, respectively. Sows were then classified as HBW or LBW if they displayed the same phenotype over successive parities, excluding data from sows whose litters represented the extremes in litter size. Cross-fostering of piglets off of HBW and LBW sows was done to standardize litters suckled to 11–15 piglets per sow. At weaning (approximately postnatal day 21), fecal swabs were taken from all sows and two female piglets per litter ($n = 90$/HBW, $n = 92$/LBW). A subset of 10 sows and a total of 20 piglets ($n = 5$ sows per phenotype, $n = 2$ female piglets per sow) from the larger population were preselected for intensive sampling. Two sampling weeks, one week apart, were chosen based on having sufficient availability of sows with minimal variation in either HBW or LBW phenotype over successive parities within those weeks and to minimize the effect of time and season. The two female piglets were selected for birth weights close to the mean of their litter of origin. At weaning, piglets were euthanized to collect intestinal digesta and tissue samples. The two female piglets selected for intensive sampling were the same piglets that had fecal swabs taken for microbial analysis.

Sampling

Female piglets were anesthetized with carbon dioxide and exsanguinated, and the digestive tract was removed via abdominal incision. Intestinal segments were clamped to avoid digesta mixing and were excised from the attached mesentery. Ileal and cecal digesta samples were snap frozen in ethanol, cooled by dry ice, and stored at −80 °C for subsequent microbial analysis. Two 3 cm^2 sections of intestinal tissue from each animal were washed in 1× phosphate-buffered saline and either fixed in 10% formalin at room temperature, or snap frozen.

Histology

Formalin-fixed ileal, jejunal, and cecal tissues were embedded in paraffin wax, and two transverse sections per pig were cut and mounted on two glass slides. Slides were stained with hematoxylin–eosin and analyzed with the use of a Nikon image analyzer. Twenty well-oriented villi and crypts were identified and measured in ileal and jejunal tissues; only crypts were measured in cecal tissue. Villus height was determined as the distance between the crypt mouth and villus tip, whereas crypt depth was defined as the invagination between two villi between the basement membrane and mouth of the crypt.

DNA extraction, 16S rRNA sequencing, and analysis

Total DNA was extracted from fecal swabs and cecal digesta using a QIAamp DNA Mini Stool Kit (Qiagen, Inc., Germantown, MD, USA). In addition to using manufacturer instructions, a bead-beating step (FastPrep instrument, MP Biomedicals, Solon, OH, USA) was added with garnet rocks to increase the DNA extraction efficiency from Gram-positive bacteria. DNA concentration was quantified using Quant-iT™ PicoGreen® dsDNA Assay Kit and subsequently diluted to 5 ng μL^{-1}. The V3–V4 region of the 16S rRNA gene was amplified using universal primers and KAPA HiFidelity Hot Start Polymerase (Kapa Biosystems, Inc. Wilmington, MA, USA). Polymerase chain reaction products were purified using AMPure XP beads (Beckman Coulter Inc., Mississauga, ON, Canada), and dual indices and Illumina sequencing adapaters attached using Nextera XT Index Kit (Illumina, Inc. Victoria, BC, CA). Second PCR products were purified, diluted to 4 nM, and pooled. The pooled PCR products were size-selected, denatured with sodium hydroxide, diluted to 4 pM in Illumina HT1 buffer, spiked with 20 PhiX, and heat denatured at 96 °C for 2 min prior to loading on the Illumina MiSeq using a MiSeq 600 cycle v3 kit (Illumina MiSeq).

Sequence data were processed using a QIIME pipeline (MacQIIME version 1.9.1 OS10.10) (Caporaso et al. 2010). PANDAseq was used for quality filtering and to assemble paired-end reads (Masella et al. 2012). Chimeras and singletons were removed using a UCHIME and UPARSE workflow with resulting sequences clustered into operational taxonomic units having >97% similarity with USEARCH (Edgar 2010, 2013; Edgar et al. 2011). Taxonomy was assigned using Ribosomal Database Project classifier V2 (Wang et al. 2007). Alpha diversity was estimated using Shannon and Chao1 indices. Alterations in microbial community structure were investigated using the Bray–Curtis dissimilarity, and analysis of similarities (ANOSIM) was used to compare the variation in species abundance and composition between LBW and HBW sows and piglets.

RNA extraction and transcriptomics

Cecal ($n = 10$), ileal ($n = 10$), and jejunum ($n = 10$) tissues were collected from piglets from HBW and LBW sows and ground into powder as described previously (Blanes et al. 2016). High-quality total RNA (RNA Integrity Number > 8) was obtained according to a previous published method (Tsoi et al. 2012) and further purified into mRNA using Dynabeads® mRNA DIRECT™ Micro Kit (Thermo Fisher Scientific, Toronto, ON, Canada). One nanogram of mRNA was used as a starting material for cDNA synthesis using SMARTer Stranded RNA-Seq Kit HT for the Illumina® Sequencing Platform (Takara Bio USA, San Francisco, CA, USA). All cDNA was qualified and quantified according to the kit manual before sending to McGill University and Genome Quebec Innovation Centre (Montreal, QC, Canada) for library quality control and Illumina HiSeq4000 PE 100 bp sequencing service. Only libraries passing the quality control test were used for RNASeq.

Sequencing reads were obtained and aligned to the pig reference genome (Sscrofa11.1) using the QSeq module of the ArrayStar software program (DNAStar version 15.0, Madison, WI, USA) to find differentially expressed (DE) genes between HBW and LBW piglets

Table 1. Biological data from litters of sows of phenotype distinguished high birth weight (HBW) and low birth weight (LBW) litters.

	HBW ($n = 46$)	LBW ($n = 45$)	P
All sows			
Birth weight (kg)	1.54 ± 0.02	1.20 ± 0.02	<0.001
Total piglets born per litter	14.6 ± 0.45	14.8 ± 0.40	0.752
Total piglets born alive per litter	13.5 ± 0.43	13.6 ± 0.44	0.844
Small intensive sampling	**($n = 5$ sows)**	**($n = 5$ sows)**	
Birth weight (kg)[a]	1.71 ± 0.11	1.26 ± 0.07	0.018
Total piglets born per litter[b]	13.4 ± 0.50	15.8 ± 0.65	0.009
Total piglets born alive per litter[b]	12.2 ± 0.84	13.4 ± 0.84	0.340
Total piglets weaned per litter[b]	10.4 ± 0.59	9.2 ± 0.59	0.189
Intestinal length (cm)[a]	327.6 ± 9.6	299.6 ± 14.2	0.404
Weaning weight (kg)[a]	6.57 ± 0.15	5.76 ± 0.59	0.377

Note: Means are considered significantly different at $P < 0.05$, with $P \geq 0.05 < 0.10$ considered a trend.

[a]Two piglets per sow were subsampled equating to $n = 10$ piglets per HBW and LBW, respectively.

[b]Total piglets averaged from each of the $n = 5$ sows per HBW and LBW phenotype.

from each gut tissue. A kilobase per million mapped reads (RPKM) normalization strategy applied according to the procedures of Mortazavi and associates (Mortazavi et al. 2008). DE genes were calculated at twofold change with 95% confidence using Student's t test with total mean RPKM equal to 1 or higher at least in one group. To correct for multiple comparisons, P values were adjusted using the Benjamini Hochberg method with a false discovery rate q value of 0.05 (Lasergene, DNAStar version 15.0, Madison, WI, USA).

Calculations and statistical analysis

Data are presented as treatment means ± standard error of the mean. A mixed model was used to compare BW and piglets born per litter, with sow used as the random variable. For data from the small intensive sampling study, a mixed model was used to compare birth and weaning weight of piglets, intestinal length, villous height, and crypt depth with litter nested within birth weight phenotype used as a random variable and date of weaning included as a covariate when significant. Non-normally distributed 16S rRNA data were analyzed using Kruskall–Wallis rank sum test. Significance was set as $P < 0.05$, with $P \geq 0.05 < 0.10$ considered a trend. Statistical analyses were done with GraphPad Prism version 6.02 (La Jolla, CA, USA), SAS (University Edition; SAS Institute Inc., Cary, NC, USA), and R (version 3.2.1).

Results

The numbers of total piglets born and total born alive (total born minus mummies and stillborns) per litter did not differ ($P > 0.05$) between HBW and LBW phenotype sows from the larger population. However, piglet BW was greater ($P < 0.05$) in piglets born from HBW versus LBW sows (Table 1). Within the subsampled group, BW was also higher ($P < 0.05$) in piglets born to HBW versus LBW sows. In the subsampled group, total piglets born per litter was higher ($P < 0.05$) in LBW sows versus HBW sows, indicating the increase in birth weight from HBW sows in this small subpopulation is potentially a function of small sample bias. However, total piglets born alive and total piglets weaned (due to cross-fostering) in the subsampled group did not differ between HBW and LBW sows ($P > 0.05$). Cross-fostering was practiced in both HBW and LBW phenotype litters, explaining the discrepancy between total born and total weaned in the subsampled group. The difference in weight seen at birth between piglets from HBW and LBW sows was not observed at weaning (21 ± 2 d of age) ($P > 0.05$). Furthermore, no distinctions in intestinal length were observed between the piglets from HBW or LBW sows ($P > 0.05$).

Histology

There were no overall differences in villous height, crypt depth, or their ratio in jejunum, ileum, and cecum of piglets from HBW or LBW sows (Table 2). However, piglet weight gain was positively correlated to jejunal crypt depth ($R = 0.583$; $P = 0.007$) and ileal crypt depth ($R = 0.413$; $P = 0.070$; Fig. 1).

Microbial composition

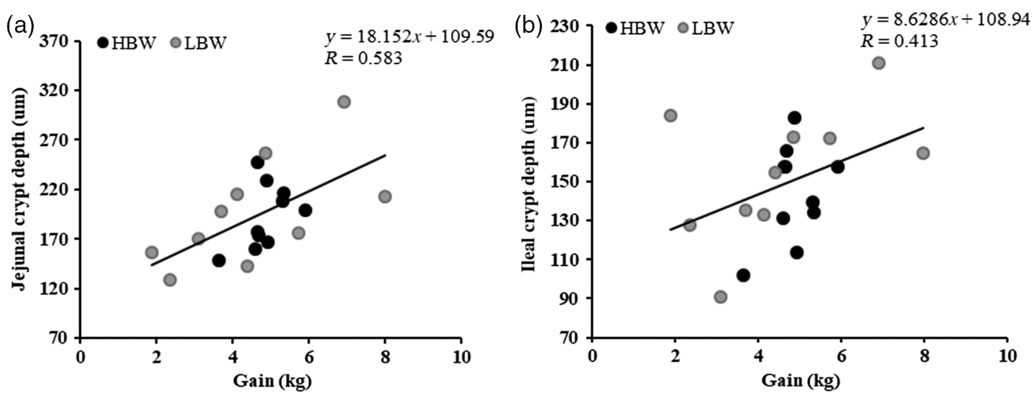
Bacterial 16S rRNA sequencing of sow feces resulted in a total of 8 193 605 sequences that were rarefied to 1000 reads per sample for alpha diversity. As expected, sows with designated LBW or HBW phenotypes did not have differing fecal microbial community structure, determined using the Bray–Curtis dissimilarity (ANOSIM; $P > 0.05$; Fig. 2). Furthermore, there were no observed differences in microbial species evenness or richness in sow feces, as determined by the Shannon and Chao1 index, respectively ($P > 0.05$; Table 3) or predominating

Table 2. Jejunum, ileum, and cecum intestinal morphology in piglets from high birth weight (HBW) and low birth weight (LBW) phenotype sows.

	HBW ($n = 10^a$)	LBW ($n = 10^a$)	SEM	P
Jejunum				
Villous height (μm)	322.1	352.4	28.3	0.956
Crypt depth (μm)	192.5	196.6	9.78	0.516
Ratio	1.69	2.00	0.20	0.932
Ileum				
Villi height (μm)	752.8	578.7	62.3	0.308
Crypt depth (μm)	144.1	154.6	6.57	0.124
Ratio	5.89	4.26	0.54	0.184
Cecum				
Crypt depth (μm)	286.3	298.3	10.4	0.532

Note: Means are considered significantly different at $P < 0.05$ with $P \geq 0.05 < 0.10$ considered a trend. SEM, pooled standard error of the mean.
[a]Two piglets per sow were subsampled for each phenotype equating to $n = 10$ piglets per HBW and LBW, respectively.

Fig. 1. Pearson's correlation between the rate of (*a*) gain and jejunal crypt depth ($R^2 = 0.34$; $P = 0.007$) and (*b*) gain and ileal crypt depth ($R^2 = 0.17$; $P = 0.070$). HBW, high birth weight; LBW, low birth weight.



taxonomy (data not shown). Sequencing piglet fecal microbiota resulted in a total of 267 692 reads that were rarefied for alpha diversity to 1000 reads per sample. Fecal microbial community structure was also not different between piglets from HBW and LBW sows, as shown by the principal coordinates analysis (PCoA) plot based on the Bray–Curtis dissimilarity (Fig. 3; ANOSIM; $P > 0.05$). Although overall microbial community structure of piglet feces did not differ, piglets from HBW sows had an increased species richness (Chao 1 index, $P = 0.006$; Table 3) and evenness (Shannon index, $P = 0.020$) in comparison to piglets from LBW sows. The most predominant taxonomic change between HBW and LBW piglets was the increase in *Proteobacteria*, specifically *Enterobacteriaceae*, in LBW piglets ($P = 0.004$; Table 4). Other changes in fecal microbial composition included increased ($P < 0.05$) relative abundance of *Bacteroidales*, *Butyricimonas*, *Fusobacterium*, *Victivallaceae*, *Desulfovibrio*, *Treponema*, and reduced ($P < 0.05$) abundance of *Odoribacter*, *Enterococcaceae*, *Streptococcus*, and *Erysipelorichaceae* in HBW versus LBW piglets.

Cecal microbiota sequencing resulted in a total of 556 783 reads that were rarefied to 8000 reads per sample for alpha diversity. Cecal microbial community structure was distinguishable between HBW and LBW piglets shown in the PCoA based on the Bray–Curtis dissimilarity (ANOSIM; $P = 0.004$; Fig. 4). The altered community structure was mainly due to increased ($P < 0.05$) *Fusobacterium* and reduced ($P < 0.05$) *Spirochaetes*, *Synergistaceae*, and *Verrucomicrobia* in HBW versus LBW piglets (Table 5). However, no differences in alpha diversity were observed in cecal digesta of HBW and LBW piglets. No changes in ileal microbial structure, composition, or species richness or evenness were detected between HBW and LBW piglets (data not shown).

Intestinal transcriptome

Only modest effects were observed in intestinal transcriptome between HBW and LBW piglets. In the jejunum of LBW piglets, genes *PROSER3*, *LOC106510122*, *CADPS*, *LOC100514515*, and Period2 (*PER2*) were

Fig. 2. Fecal microbial community structure between high birth weight (HBW-S) and low birth weight (LBW-S) phenotype sows. Determined using Bray–Curtis dissimilarity; no differences were observed in sow fecal microbial composition (analysis of similarities; $P > 0.05$; $n = 42$/group).

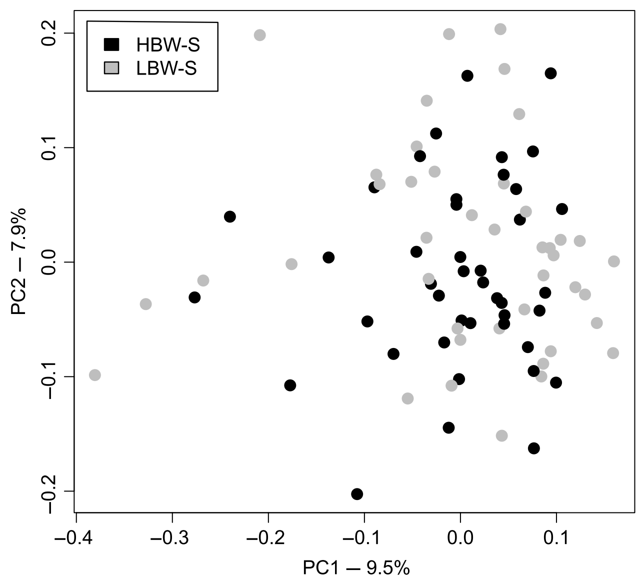


Fig. 3. Fecal microbial community structure between piglets born from distinguished high birth weight (HBW) and low birth weight (LBW) phenotype sows. Determined using Bray–Curtis dissimilarity; no differences were observed in piglet fecal microbial structure (analysis of similarities; $P > 0.05$; $n = 76$ HBW; $n = 81$ LBW).

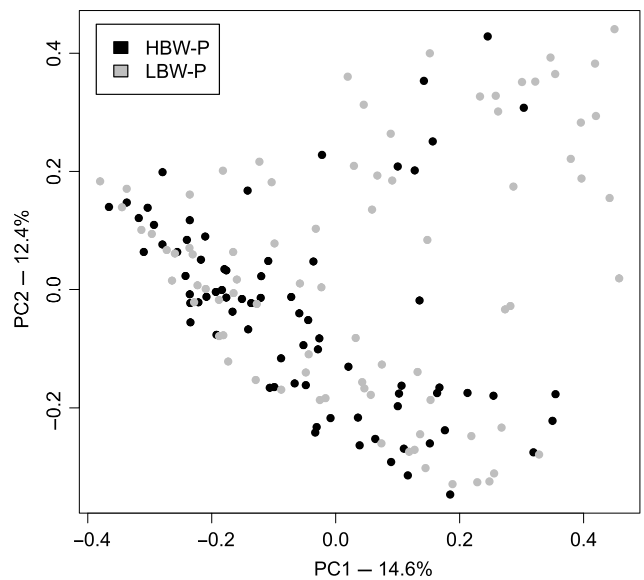


Table 3. Species richness and evenness of sows and piglets from phenotype distinguished high birth weight (HBW) and low birth weight (LBW) litters.

	HBW	LBW	SEM	*P*
Sow feces	**($n = 42$)**	**($n = 42$)**		
Shannon index	6.60	6.50	0.07	0.331
Chao1 index	377.5	362.2	7.50	0.227
Piglet feces	**($n = 76$)**	**($n = 81$)**		
Shannon index	4.98	4.60	0.07	0.020
Chao1 index	162.5	142.6	4.09	0.006
Piglet cecal digesta	**($n = 10^a$)**	**($n = 10^a$)**		
Shannon index	5.46	5.72	0.14	0.384
Chao1 index	307.2	289.7	13.26	0.602

Note: Means are considered significantly different at $P < 0.05$ with $P \geq 0.05 < 0.10$ considered a trend. SEM, pooled standard error of the mean.

[a]Two piglets per sow were sub-sampled for each phenotype equating to $n = 10$ piglets per HBW and LBW, respectively.

upregulated compared with HBW piglets (Table 6; $P < 0.001$). In the ileum of LBW piglets, genes *ELAVL4* and Klech-like family member 38 (*KLHL38*) were upregulated ($P < 0.05$), and carboxylesterase1 (*CES1*) and *IL1B* tended to be downregulated ($P < 0.10$). In the cecum, where microbial community structure was altered, hexokinase2 (*HK2*), interferon regulatory factor 8 (*IRF8*), *TNIP3*, and *PLA2G2D* were upregulated ($P < 0.05$), and *OTC* and *PCP4L1* were down regulated ($P < 0.05$) in LBW piglets.

Discussion

In the commercial swine industry, some sows have a recurring high or low average litter birth weight phenotype (Smit et al. 2013). Due to the potential economic consequences of sows consistently having LBW litters, the objectives of this research were to determine if piglets from LBW sows have altered pre-weaning growth performance, gut-associated microbiota, intestinal architecture, and transcriptome at weaning.

Growth performance

In the current study, piglets from LBW sows had significantly lighter birth weights than piglets from HBW sows. The discrepancy in piglet birth weights between LBW and HBW sows is hypothesized to be a consequence of early intrauterine crowding driven by indirect genetic selection for high ovulation rates resulting in excessive intra-uterine crowding of embryos in early gestation, limitations on placental development and ultimately IUGR of fetuses at term (Foxcroft et al. 2009). Although total piglets born per litter were higher in the subsampled LBW sows, total piglets born alive were equivalent to HBW sows, indicating LBW sows have reduced piglet viability. However, by weaning (day 21), piglets from LBW sows no longer differed in body weight compared with piglets from HBW sows. These results are contradictory to previous work by Smit et al. (2013), which found LBW piglets had decreased postnatal growth during all phases of production. However, in the current study due to cross-fostering, sows had on average one to two less piglets suckling per litter in both

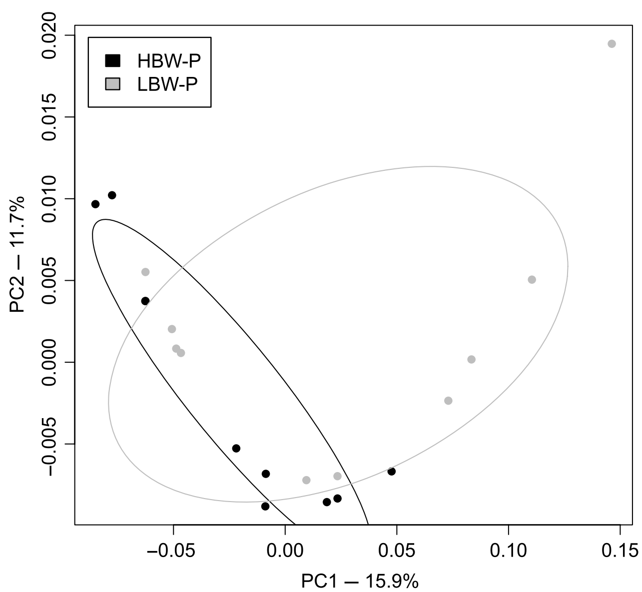
Table 4. Relative abundance of predominating phyla and changed taxonomy in piglet feces from phenotype high birth weight (HBW) and low birth weight (LBW) sows.

Phyla	HBW ($n = 76$)	LBW ($n = 81$)	SEM	P value
Actinobacteria	0.67	0.95	0.11	0.053
Bacteroidetes	35.88	31.35	1.17	0.062
Firmicutes	40.50	38.96	1.12	0.287
Fusobacteria	0.81	0.29	0.13	0.024
Lentisphaerae	0.21	0.16	0.03	0.093
Planctomycetes	0.13	0.10	0.02	0.712
Proteobacteria	12.37	20.93	1.34	0.003
Spirochaetes	4.20	2.75	0.39	0.001
Synergistetes	3.14	2.76	0.39	0.047
Tenericutes	0.14	0.14	0.03	0.787
Verrucomicrobia	1.41	1.27	0.24	0.762
Changing taxonomy				
Bacteroidetes				
o__*Bacteroidales*	5.09	3.72	0.43	0.026
g__*Butyricimonas*	1.62	1.29	0.14	0.006
g__*Odoribacter*	0.47	0.61	0.13	0.026
f__[*Paraprevotellaceae*]	0.42	0.35	0.07	0.073
Firmicutes				
f__*Enterococcaceae*	0.12	0.40	0.07	0.024
g__*Streptococcus*	0.28	0.69	0.06	0.005
f__*Christensenellaceae*	2.23	1.61	0.16	0.052
g__*Coprococcus*	0.54	0.48	0.07	0.052
g__*Roseburia*	0.30	0.27	0.06	0.070
f__*Ruminococcaceae*	10.68	9.37	0.55	0.066
g__*Ruminococcus*	2.08	1.79	0.15	0.062
g__*Phascolarctobacterium*	1.26	1.06	0.07	0.098
f__*Erysipelotrichaceae*	0.18	0.25	0.03	0.031
Fusobacteria				
g__*Fusobacterium*	0.81	0.29	0.11	0.028
Lentisphaerae				
f__*Victivallaceae*	0.18	0.13	0.03	0.023
Proteobacteria				
g__*Sutterella*	0.86	0.52	0.08	0.047
g__*Desulfovibrio*	0.71	0.50	0.05	0.011
g__*Helicobacter*	1.04	0.69	0.14	0.052
f__*Enterobacteriaceae*	7.28	15.56	1.30	0.004
Spirochaetes				
g__*Treponema*	3.85	2.35	0.37	0.004
Verrucomicrobia				
f__[*Cerasicoccaceae*]	0.15	0.27	0.07	0.069

Note: Means are considered significantly different at $P < 0.05$ with $P \geq 0.05 < 0.10$ considered a trend. SEM, pooled standard error of the mean; o_, order; g_, genus; f_, family.

phenotype groups, potentially resulting in sufficient milk production allowing piglets to express their full growth potential. Furthermore, it has been suggested that limitations in growth performance of piglets from LBW sows will only be measurable in the grower and finisher phases of production, possibly explaining why no differences were observed at weaning in the current study (Foxcroft et al. 2007). In support of this, previous research has shown piglets from LBW sows take an additional 9 d to reach market weight (Smit et al. 2013). Consistent with previous literature, LBW piglets had a 9% reduction in small intestinal length, (Michiels et al. 2013; Huygelen et al. 2015). Although not significantly reduced due to limited animal numbers, the 9% reduction in the intestinal length of LBW piglets still has the potential to change digestive function by reducing absorptive surface area, possibly impacting feed efficiency later in life. The small sample size of the subsampled piglets ($n = 10$ piglets from $n = 5$ sows per phenotype) is a limiting factor that may have affected the ability to detect differences in growth performance.

Fig. 4. Cecal microbial community structure between piglets born from distinguished high birth weight (HBW) and low birth weight (LBW) phenotype sows. Cecal microbial community composition was significantly separated between piglets from HBW and LBW sows, as determined using the Bray–Curtis dissimilarity (analysis of similarities; $P = 0.004$; $n = 10$/HBW and LBW).



Histology

A major disadvantage observed in LBW piglets is delayed small intestinal development, reduced villous area (D'Inca et al. 2011), and reduced mucosal height (Alvarenga et al. 2013). Delays in intestinal development at birth can increase the risk for pathogen colonization or reduced digestive efficiency (Wang et al. 2010). Contradictory to previous findings by Alvarenga et al. (2012) who found duodenal mucosal height reduced in LBW piglets, the intestinal structure remained unchanged between LBW and HBW piglets at weaning. In the current study, HBW and LBW piglets were reared in equivalent sized litters, and as a result, weaning weight did not differ between groups, suggesting compensatory gain took place in LBW piglets. Furthermore, a positive correlation between weight gain and crypt depth indicates a potential compensatory mechanism occurring in LBW piglets. In agreement with our results, it has been observed that LBW piglets have increased average daily gain accompanied by significantly deeper crypts compared with normal birth weight counterparts (De Vos et al. 2014). Crypt cell proliferation is a common compensatory function observed in humans following a loss of functional small bowel surface area (Rubin and Levin 2016).

Microbial composition

Gut-associated microbiota are established immediately after birth, characteristically from the sow and surrounding environment. Beacuse LBW and HBW phenotype sows came from the same environment, it is not surprising that they have indistinguishable fecal microbial community structures. Remarkably, the cecal microbial community structure was altered in conjunction with reduced fecal species richness and evenness in LBW piglets. The most striking difference in fecal microbial composition between LBW and HBW piglets was a 114% increase in *Enterobacteriaceae* in LBW piglets. This characteristic bloom of *Enterobacteriaceae* and reduced microbial diversity has been extensively characterized, and it is frequently observed in LBW human infants (Unger et al. 2014). The initial microbial colonization and successional patterns in neonates is a window of opportunity for proper immune education and development. Because of the flexibility and instability of neonatal microbiota, it is vulnerable to deviations, such as this bloom of *Enterobacteriaceae*, which may have immune consequences later in life. It has been established that gut microbiota composition can alter disease susceptibility (Thiemann et al. 2017), with lipopolysaccharide (LPS) exposure from specific bacteria, such as *Escherichia coli*, fundamental in immune system development and education (Vatanen et al. 2016). In the current study, LBW piglets also tended to have reduced fecal *Bacteroidetes* and *Ruminococcaceae*, and reduced cecal *Bacteroides*, in agreement with previous work (Morissette et al. 2017). Members of the *Ruminococcaceae* family are cellulolytic short-chain fatty acid (SCFA) producers and have been associated with reduced predicted *Salmonella* shedding in pigs (Bearson et al. 2013).

Other notable commensals reduced in cecal digesta of LBW piglets included *Roseburia*, *Faecalibacterium*, *Dialister*, and *Veillonella*. Fiber-degrading bacteria such as these also utilize polysaccharides that are indigestible to the host, producing SCFA. Members of *Faecalibacterium* and *Roseburia* are known butyrate producers, the preferential energy substrate for colonocytes (Pryde et al. 2002). Reduction of these butyrate-producing commensals may have consequences for colonic barrier and immune function. The reduction in fiber-degrading bacteria may also explain the reduced feed efficiency and increased days to market observed previously, as the SCFAs can be used as energy substrates by the pig.

Transcriptomics

Although only modest changes in intestinal transcriptome were observed, most DE genes belonged to either metabolic or immune pathways, suggesting there may be further consequences for LBW piglets. Among the genes upregulated in LBW piglets was *PER2*. *PER2* is interesting as it is a component of the circadian clock set of genes and is expressed in a daily pattern (Pan and Hussain 2009). It has been established that gut microbiota and the circadian clock are highly connected, with microbiota-depleted mice having increased *PER2* expression in ileum and colon (Mukherji et al. 2013). Likewise, LBW piglets in the current study had reduced fecal

Table 5. Relative abundance of predominating phyla and changed taxonomy in piglet cecal digesta from phenotype high birth weight (HBW) and low birth weight (LBW) sows.

Phyla	HBW ($n = 10^a$)	LBW ($n = 10^a$)	SEM	*P*
Bacteroidetes	53.26	47.29	1.44	0.174
Cyanobacteria	0.35	0.20	0.07	0.151
Elusimicrobia	0.14	0.13	0.05	0.937
Firmicutes	25.66	30.01	1.84	0.290
Fusobacteria	7.51	0.87	1.66	0.023
Lentisphaerae	0.49	0.34	0.10	0.880
Proteobacteria	6.90	5.24	0.81	0.199
Spirochaetes	2.30	6.86	0.87	0.010
Synergistetes	1.12	2.74	0.43	0.070
Tenericutes	0.15	0.21	0.04	0.273
Verrucomicrobia	1.23	5.19	1.22	0.034
Changing taxonomy				
Bacteroidetes				
o__*Bacteroidales*	4.54	7.29	0.81	0.049
g__*Odoribacter*	1.52	0.50	0.35	0.041
g__*Bacteroides*	9.36	5.56	1.47	0.034
Firmicutes				
f__*Lachnospiraceae*;g__	1.39	2.31	0.45	0.023
g__*Roseburia*	0.53	0.12	0.10	0.008
g__*Faecalibacterium*	0.09	0.01	0.02	0.009
g__*Oscillospira*	2.05	4.04	0.49	0.034
g__*Dialister*	0.06	0.00	0.02	0.013
g__*Veillonella*	0.26	0.09	0.04	0.019
f__*Erysipelotrichaceae*;g__RFN20	0.31	1.06	0.17	0.034
Fusobacteria				
g__*Fusobacterium*	7.50	0.86	1.66	0.023
Proteobacteria				
g__*Bilophila*	0.05	0.28	0.04	0.001
g__*Actinobacillus*	2.16	0.83	0.32	0.023
Spirochaetes				
g__*Sphaerocheata*	1.49	5.62	0.85	0.019
Synergistetes				
f__*Synergistaceae*;Other	0.01	0.30	0.10	0.009

Note: Means are considered significantly different at $P < 0.05$ with $P \geq 0.05 < 0.10$ considered a trend. SEM, pooled standard error of the mean; o_, order; g_, genus; f_, family.

[a]Two piglets per sow were subsampled for each phenotype equating to $n = 10$ piglets per HBW and LBW, respectively.

microbial richness indicated by a lower Chao1 index. Disruption of clock genes has been associated with metabolic disorders, with *PER2* necessary to control proper lipid metabolism (Grimaldi et al. 2010). Another metabolic gene found downregulated in LBW piglets was *CES1*. *CES1* has been shown to be increased in adipocytes of obese versus lean subjects (Benabdelkamel et al. 2015). Although at time of sampling, LBW animals did not differ from HBW counterparts, they originally had significantly lower birth weights, and they may have altered adiposity versus their HBW counterparts. Coinciding, the *KLHL38* was found to be upregulated in ileum of LBW piglets. The *KLHL38* gene has been proposed to play a role in preadipocyte differentiation in the chicken (Zhang et al. 2017). Along with altered metabolic genes, *HK2* was also DE. *HK2* phosphorylates glucose to produce glucose-6-phosphate, and it is the first step in aerobic glucose metabolism. Interestingly, LBW piglets had increased fecal *Enterobacteriaceae*, mainly facultative anaerobes. Expansion of facultative anaerobes typically indicates a microaerophilic intestinal environment, potentially explaining the increased *HK2* expression required for aerobic respiration. Additionally, *HK2* expression has shown to be increased with bacterial and protozoan infections that are often accompanied by increased oxygen content and aerobic respiration (Archambaud et al. 2013; Menendez et al. 2015).

Several immune genes were also DE between LBW and HBW piglets including the pro-inflammatory cytokine, interleukin 1 beta (IL-1β). IL-1β promotes activation and functions of dendritic cells, macrophages, and neutrophils essential for the host response against pathogens.

Table 6. Differentially expressed genes in jejunum, ileum, and cecum of piglets from high birth weight (HBW) versus low birth weight (LBW) litters.

Gene	HBW ($n = 10^a$)	LBW ($n = 10^a$)	SEM	*P*	Fold change
Jejunum					
PROSER3	1.01	2.24	0.055	0.003	2.2
LOC106510122	0.43	1.17	0.071	0.004	2.7
CADPS	0.52	1.09	0.038	<0.001	2.2
LOC100514515	0.54	1.09	0.033	<0.001	2.0
PER2	0.45	1.03	0.071	0.006	2.3
Ileum					
CES1	170.9	73.9	6.240	0.052	−2.3
IL1B	1.89	0.72	0.074	0.053	−2.6
ELAVL4	1.13	2.31	0.075	0.046	2.0
KLHL38	0.78	1.82	0.066	0.043	2.3
Cecum					
HK2	3.98	8.91	0.314	0.004	2.2
IRF8	2.90	6.78	0.197	0.004	2.3
TNIP3	1.83	6.15	0.310	0.007	3.4
OTC	5.21	2.42	0.200	0.009	−2.2
PCP4L1	5.27	2.22	0.167	0.001	−2.4
PLA2G2D	4.17	11.54	0.453	0.006	2.8

Note: Means are considered significantly different at $P < 0.05$ with $P \geq 0.05 < 0.10$ considered a trend. SEM, pooled standard error of the mean; *PER2*, Period2; *CES1*, carboxylesterase1; *KLHL38*, Klech-like family member 38; *HK2*, hexokinase2; *IRF8*, interferon regulatory factor 8.

[a]Two piglets per sow were subsampled for each phenotype equating to $n = 10$ piglets per HBW and LBW, respectively.

Reduced IL-1β expression in LBW animals may indicate altered immune system function (Lopez-Castejon and Brough 2011). Further evidence suggesting LBW piglets have an altered immune status is observed in their upregulated *IRF8*. The *IRF8* gene is a transcription factor regulating myeloid cell maturation and differentiation, and is proposed to play a critical role against intracellular pathogens (Dror et al. 2007; Salem and Gros 2013), which also may be linked to the bloom of *Enterobacteriaceae* in LBW piglets. TNFA-interacting protein 3 was also upregulated in LBW piglets, and it is known to regulate NF-κB activation in monocytes in response to TNFκ, LPS, and IL-10. Interestingly TNFAIP3 has been associated with reductions in IL-1β expression observed in the ileum of LBW piglets (Abbasi et al. 2014). Taken together, the reduced IL-1β and increased *IRF8* and TNFAIP3 indicate LBW piglets may be responding to the expanded *Enterobacteriaceae* population. The fact piglets were sampled immediately at weaning and did not undergo the typical stress encountered with weaning may be the reason only subtle changes in the intestinal transcriptome were observed.

Conclusion

In the current study, it was found that piglets originating from LBW sows had altered fecal and cecal microbial structures that included a large bloom of *Enterobacteriaceae* and corresponding upregulated immune genes. Due to the strong relationship between gut-associated microbiota and immune programming, the microbial shifts observed in LBW piglets may have long-lasting immune consequences that may divert energy away from growth and reduce efficiency later in life. Additional research is necessary to determine how LBW piglet performance is affected later in life or in a challenged environment, to fully understand why such large performance discrepancies are observed in other datasets.

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